Qian Ouyang

qouyang@nd.edu | cv.qianoy.uk | github.com/Owen-Ou-Yang

Education

Xi'an Jiaotong University 2023 - 2027 (expected)
Undergraduate studies in Artificial Intelligence
Artificial Intelligence Experimental Class

University of Notre Dame August - December 2025
Undergraduate Exchange Student (non-degree)
Semester GPA: 3.917/4.0 | 12 credits
Selected coursework: Computer Vision - graduate level (A); Natural Language Processing (A); Computer Graphics (A).

Research Experience

University of Notre Dame

Research Assistant October 2025 - Present
Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

Machine learning for polymer simulation

Compared preliminary polymer-density estimates from MACE simulations with reference data. Investigated GPU-memory limitations in UMA / FAIR-Chem calculations and compared runtime, GPU memory use, and GPU-hours across selected MACE runs.

Representation learning for molecular dynamics

Studied JEPA-based representations for coarse-grained polymers and atomistic water. In the water experiments, used pretrained MACE features and atomic velocities to predict short-time atomic displacements, and compared temporal JEPA training with direct supervision.

Summer Research Participant (iSURE) July - September 2026

Course Project

Driving Coach: Lane and Traffic-Object Detection Fall 2025
Computer Vision, University of Notre Dame | Instructor: Prof. Adam Czajka

Adapted an existing YOLOv1-style detector with a pretrained ResNet-18 backbone for lane and traffic-object detection in BDD100K driving scenes. Used an unseen Shenzhen highway video for qualitative evaluation and analyzed errors involving reflections, lane geometry, and traffic signs.

Skills

Programming: Python, C++